The Limitations of Generative AI

What these systems cannot (yet) do, and why

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The problem with how AI is explained

- Public understanding is built on demonstrations of success: successes are easy to see, the structure of failures is not.
- Someone who has only seen a system succeed cannot tell a reliable tool from an unreliable one, judge announced fixes, or assess breathless claims in either direction.
- This talk supplies the missing half of the picture, under one discipline: every claim is marked as **observed** (measured), **explained** (well-argued mechanism), or **narrated** (repeated until it sounds established).
- The bibliography was not assembled from memory: all 51 entries were verified against DBLP, the arXiv API, and Crossref by a script, and open-access copies of 29 cited papers are archived with pinned hashes. Two claims that failed verification were dropped, not cited.

Three families of failure — not one laundry list

Family	Signature	Matching remedy
Statistical	errors average out over samples	better data, better measurement
Dynamical	errors circulate through feedback	separate learning from acting
Normative	the violated rule cannot even be represented	inspectable structure outside the weights

The consequential category error

Offering a statistical remedy for a dynamical or normative failure — the most common shape of announced fixes and AI policy debates alike.

Statistical failures

Confabulation is the training objective working as designed

- "Hallucination" suggests a perception problem; the mechanism is the objective itself rewarding plausible-sounding continuation.
- Observed: scaling *dilutes* but does not remove it; retrieval patches the symptom on popular topics while the long tail stays exposed (Mallen et al., 2023).
- The state is *malleable*: a masked-word test that changes one word of the prompt flips the answer it gives (Pride & Prejudice marriage dates, .org/.com) — what is "known" versus what is reported come apart.

Context and calibration

Context is a budget, not a capability

- Effective vs. advertised length: measured gap (Hsieh et al., 2024, RULER); “lost in the middle”: recall is high at the edges and collapses in between (Liu et al., 2024).
- Retrieval augmentation inherits the problem: whatever is retrieved poorly is silently absent.

Calibration

- A system that is right 80% of the time should say so; badly calibrated systems sound equally sure either way.
- Masked-word experiments show the disconnect between held knowledge and reported confidence directly.

Even the measurement layer is compromised

- **Contamination**: test questions leak into training data — examinations become memory tests.
- Fresh GSM1k problems matching the GSM8k distribution: drops of up to 8 percentage points in several model families — though, honestly, the frontier systems showed little of it (Zhang et al., 2024).
- So a headline benchmark number can conflate reasoning with memorization, and you cannot tell which from the score alone.

Dynamical failures

Memory: the read side of state

- A fixed context window meets unbounded interaction: long sessions degrade retrieval of their own beginnings — forgetting what was said *in this conversation*.
- This is architectural, not a data problem: no amount of better training data changes the shape.
- Remedies that actually match: explicit external state — scratchpads, retrieval over interaction history, persistent stores — with write policies that are themselves inspectable.

Learning in the loop: when errors circulate

- A system that learns while it acts turns isolated errors into oscillations: feedback loops, reward hacking, objective gaming — and *model collapse* when models train on their own outputs and the tail of the distribution decays.
- The remedy is **architectural separation**: learning on one timescale, acting on another — two-process designs, frozen weights with an external controller.
- Statistical remedies (more data, better prompts) do not touch this family at all.

Normative failures

Some obligations are rules, not probabilities

- Privacy (contextual integrity, Nissenbaum 2009) is a system of norms: who may transmit what about whom, to whom, in which context.
- Formalized, it becomes a deontic logic with temporal operators (Barth et al., 2006) — requiring recipient beliefs, role and group structure, retention over time, leak probabilities.
- Read the machinery off the formalization: multi-type, multi-agent, multimodal logic with temporality and probabilism.
- Nothing in a corpus gradient can represent "this transmission violates this norm in this context."

Instances are not coherence

- A fine-tuned model sometimes refuses to repeat someone's address: unreliably, with no audit trail, no consistency across time, and no correction short of retraining.
- Coherence — norm consistency, auditability, reversibility — requires **inspectable structure outside the trained weights**.
- The semiotic layer matters here: an *indexical* link (this photo *is* this person) is a doxxing vector; an iconic rendering is representational harm; a symbolic misuse violates an identifiable norm.

Deployment reality
Narrow hybrids (guardrails, constrained decoding) are commercial. Flagship neurosymbolic systems (AlphaGeometry, AlphaProof) are research stage. The full governance-demanding profile is deployed nowhere — the vacuum is the finding.

Alternatives in the design space

Diffusion language models change the trade-offs

- Generate all positions in parallel, in any order; every token is conditioned on every other (bidirectional context).
- Observed: an 8B diffusion LM is competitive with an equivalently sized autoregressive model (LLaDA vs. LLaMA 3 8B).
- Caveats: unlearning pathological associations is harder when all tokens are mutually conditioned; latency wins come at scale.
- Scope: helps *statistical* limits (all-position access for context, parallel generation) — not the dynamical or normative families.

Delivery, incentives, and misrepresentation

The shape of the machine shapes the science

- Weights withheld, service only: transparency scores are lowest exactly where a service model hides things — training data, compute, hardware (Bommasani et al., 2023, FMTI).
- Hosted models are updated or retired without documentation, and published findings silently break (Pozzobon et al., 2023).
- Access to scale was concentrating before the boom (Ahmed, 2020); ML-based science has a reproducibility crisis of its own (Kapoor & Narayanan, 2023). Google's 2023 memo said the quiet part: no moat, and neither does OpenAI.

Why states now treat hosted AI as a dependency question

- A hosted service runs on someone else's infrastructure and jurisdiction, and the history keeps arriving: Merkel's phone (2013), Crypto AG secretly CIA-owned (2020), Danish cables used against European politicians (2021), a nine-year campaign of surveillance and intimidation against ICC staff (2024), ECHELON, and *Schrems II* (2020).
- The lesson generalizes: exposure is a property of who operates the infrastructure, not of who holds which alliance today.
- Hence *digitale Souveränität*: procurement moves (Schleswig-Holstein, Denmark) and the Data Act's switching and portability obligations.

The everyday side: what does the customer possess?

- A self-contained product keeps working when its maker does not; a service that shuts down — acquisition, repricing, lost interest, bankruptcy — takes its function with it.
- Not distinctive to AI: the standard lifecycle risk of hosting — which is exactly what the sovereignty programs legislate against.
- Self-hosting and open weights exist in the design space (the Mowgli stack is a small, deliberately modest existence proof); the question is why the dominant presentation is the one where the customer owns least.

Misrepresentation needs no dishonesty

- Each link is individually reasonable; the chain is not: vendors fund demonstrations of success, the visible scoreboard is run with the firms it ranks, and the products are tuned for approval.
- **Sycophancy** (Sharma et al., 2023): agreement with the user's stated belief across every assistant studied, traced to human preference data and amplified by preference-model optimization — apparent agreement is not evidence of correctness.
- **Leaderboard Illusion** (Singh et al., 2025): up to 27 private variants tested in parallel by select providers; 205 of 243 public models silently deprecated; restricted tuning worth up to 112% on a downstream benchmark.

Honesty about the boundary

Measured
The capability claims are shaped by the same incentives that selected the architecture: this half is evidenced.

Open, falsifiable
Whether those incentives *reduce* total progress or merely *redistribute* it toward the measurable and saleable. *Weakening conditions*: independent instruments anchoring public perception; replication failure of the asymmetries; their absence in deployed systems.

Open questions

What remains open

- Is confabulation reducible to acceptable levels under an objective that rewards plausibility? (*unresolved*)
- Do effective and advertised context lengths converge? (*persistent across generations*)
- Is reliable compositionality reachable by training alone? (*evidence points to structural support*)
- Do the incentives slow progress, or redistribute it? (*falsifiable; conditions stated*)

The discipline
In-principle limits and in-practice shortfalls are different claims, and conflating them is how both hype and dismissal manufacture their certainty.

Takeaways

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1. Ask **which family** of failure you are looking at — the remedy only exists in the matching family.
2. Distrust single-number evaluations: fresh test sets, independent instruments, style-sensitivity checks.
3. For deployed decisions, ask what you **possess**: weights, audit rights, switching paths — or only a subscription.
4. Coherence (privacy, audit, consistency) demands inspectable structure; a probability distribution cannot break a rule it cannot represent.

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- Paper (with verified bibliography and corrections log): `https://nadiayvette.frama.io/gen-ai-limits/main.pdf`
- Source on all mirrors; framagit is canonical: `https://framagit.org/NadiaYvette/gen-ai-limits`

Thank you.